

Capstone Project: Experimental Design with Real-world Datasets From Project-1

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INF2178: Experimental Design for Data Science

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IMPORTANT NOTE: Please note that certain values in the report may not align with those presented in the Colab notebook. This discrepancy is due to the fact that the values were updated after running the code multiple times to ensure accuracy prior to submission. The report, however, was based on data from an earlier execution of the code.

Report Page Count (Excluding Cover Page): 24 pages

Appendix Page Count: 9 Pages

1. Introduction

The project chosen (i.e. Project-1) for this report investigates whether monetary incentives provided to Community-Based Volunteers (CBVs) in Zambia influence maternal healthcare engagement, with a focus on two primary outcomes: (1) the number of women referred or accompanied to antenatal care (ANC) services by CBVs, and (2) the number of women who ultimately seek early ANC at healthcare facilities. To address these questions rigorously, we used both Parallel Designs and Matched Pair Designs, each applied to two separate hypotheses at the NHC-level and the facility-level. Multiple two-way ANOVAs (with and without log-transformations) were then performed, incorporating relevant blocking factors (i.e. facility type, region) and continuous covariates (i.e. distance to the District Medical Office, number of maternity beds) to isolate the impact of incentives from other facility-specific or demographic variations.

Project-1 Description

Project 1 tested whether paying small monetary incentives to community-based volunteers (CBVs) in rural Zombina would boost early antenatal care (ANC) attendance and facility deliveries. In a cluster randomized trial, 98 rural health facilities in Central Province were paired based on distance to the district medical office, number of maternity beds, and catchment-area population, and then randomized so that one in each pair provided incentives (treatment) while the other did not (control).

The study's key dependent variables at the facility level were the proportion of women attending ANC in the first trimester and the proportion of facility deliveries, while at the Neighborhood Health Committee (NHC) level researchers measured how many women were referred or accompanied to first-trimester ANC. The main factor was whether CBVs received cash incentives for each woman attending ANC before 14 weeks. Covariates included facility characteristics (i.e. distance, bed capacity, catchment population) and community characteristics (i.e. proportion of female volunteers, training group). Matching these facilities improved baseline balance and statistical power but added complexity, especially regarding risk of contamination and limitations on the generalizability of the findings. Overall, the matched-pair, parallel design aimed to control for facility differences and thus more accurately assess how incentives to CBVs might affect early ANC and delivery outcomes.

Project-2 Description

Project 2 aimed to study if providing supplementary drinking water could improve hydration and cognitive performance among rural Malian schoolchildren. Researchers used a crossover design, meaning each child was tested on two separate days: on one (i.e. the "water day") they received a refillable 1.5-liter bottle of water, and on the other (i.e. the "control day") they did not. Hydration was checked through urine specific gravity, urine color, and self-reported thirst measured both in the morning and afternoon, while cognitive tests included paper based tasks for attention (i.e. Letter Cancellation, Direct Image Difference), visual memory (i.e. Indirect Image Difference), short-term memory (i.e. Forward and Reverse Digit Recall), and visuomotor skills (i.e. Line Tracing).

To control for individual differences, each pupil acted as their own control, but this also introduced a practice effect (where children's scores improved simply because they repeated the tests). Other covariates such as sex, grade level, morning drinking/eating habits, and the order of testing were also tracked to spot any confounding factors. Ultimately, while extra water did lower afternoon urine specific gravity (i.e. indicating better hydration), the learning effect over repeated tests made it difficult to see a clear benefit on cognitive scores; thus, the crossover design was helpful in reducing between child differences, but also complicated the ability to isolate water's true effect on cognition due to repeated testing.

Project-1 Data Cleaning & Transformation Process

To reduce redundancy between the NHC-level and Facility-level datasets and prepare them for parallel and matched-pair analyses using Pandas in Python, we first filtered the data to include only the columns required for the two experiments, thereby preventing unnecessary overhead and keeping the workflow clear. We removed rows with null values to maintain data integrity and avoid complications during transformations or statistical operations. Next, in accordance with the data dictionary, we converted the *treatment*, *fid*, *pair*, *facility type*, and *training* variables to categorical, ensuring efficient memory usage and accurate grouping for comparisons. We then restricted the dataset to January through June of 2020 because, according to the study, monetary incentives were assigned to community-based volunteers in treatment sites during this specific period, and they received payments for each woman referred or accompanied in the first trimester of pregnancy within that time frame. Finally, we joined the NHC-level and Facility-level datasets on *fid* and *month*, ensuring that corresponding observations were properly aligned for further comparative analyses.

2. Parallel Design

In our design, participants are randomly assigned to either a treatment group or a control group at the start of the study, and they remain in their assigned group throughout the entire observation period. To achieve medium effect, a total of 146 participants are selected from the merged data. Splitting the participants into either treatment or control. Random participants will be selected from the original dataset such that:

- Group 1 (Treatment - Incentives, 73 participants)
- Group 2 (Control - No Incentives, 73 participants)

We selected the parallel design because our intervention (monetary incentives for CBVs) is a between-group factor rather than rotating between participants. Participants are either in the treatment group or in the control group, but not both. Parallel design ([Figure 1](#)) is well suited here when outcomes are compared across independent groups. In our case, once a facility is assigned a treatment condition, it remains in that condition throughout the study period.

Therefore, applying a parallel design here could reflect a real-world setting, where switching treatments is not feasible.

We used different hypotheses for Hypothesis 1 and Hypothesis 2. Although both hypotheses evaluate the impact of monetary incentives on women's access to ANC services, they are tested at different levels of analysis and capture distinct behavioral outcomes. Hypothesis 1 focuses on CBV behavior (whether incentives lead to more referrals or accompaniment of women to health facilities). This reflects the output of the CBV directly and is measured using the total number of women referred/accompanied. Hypothesis 2 focuses on the facility outcome (whether incentives

to CBVs ultimately increase the number of women arriving for ANC in the first trimester). This outcome is indirectly tied to CBV behavior, but also depends on facility accessibility, readiness, and women's follow-through.

1.1 Hypothesis 1 NHC-Level From Merged DataFrame

Null Hypothesis (H_0): The intervention/treatment (incentives for CBVs) will lead to an increase in the total number of women referred and accompanied by CBVs (Community-Based Volunteers).

Alternative Hypothesis (H_a): The intervention/treatment (incentives for CBVs) will NOT lead to an increase in the total number of women referred and accompanied by CBVs (Community-Based Volunteers).

Subset Data Description:

Variable Name	Variable Type	Description	Why It Was Chosen
treatment_nhc	Categorical (0 = control, 1 = treatment)	Indicates whether the NHC received financial incentives	Independent variable
total	Continuous (Integer)	Total number of women referred or accompanied by CBVs in each NHC	Dependent variable , reflecting combined outreach activities.
fid	Categorical	Unique identifier for each facility or NHC	Used to group datasets across levels.
month	Categorical	Month of observation	Used to control for seasonal variation
Group_Label	Categorical	Label for treatment/control group in some merged datasets (e.g., "Treatment")	For tracking treatment status in merged/resaped views
facilitytype_nhc	Categorical	Type of facility associated with each NHC (e.g., rural, hospital)	Used as a blocking variable to account for the variability within different types of facilities.
femalenhc_ratio	Continuous (Float)	Proportion of women in the NHC	Used as the covariate capturing potential gender influence on referral and outreach.

Parallel Design Setup with Blocking Factor for Hypothesis 1

Objective: To determine whether offering financial incentives to community-based volunteers (CBVs) leads to a higher number of women referred or accompanied to health facilities compared to a control group with no incentives, while controlling for potential seasonal effects using facility_nhc as a blocking factor.

Setup:

- **Blocking Factor:** facility_nhc is chosen as the blocking factor since the outcome variable (i.e. total referrals/accompanied number) could be dependent on the type of facility that the womens are being placed in. We want to account for the variability within different types of facilities. [Figure 2](#) illustrates the distribution of the dependent variable with the blocking factor incorporated.

- **Randomization:** Participants are randomly assigned to one of two groups. [Figure 3](#) and [figure 4](#) illustrate the distribution for the total number of referrals and accompaniments as a barchart and violin plot from the selected subset of participants. These two figures are also used to check for normality of the dependent variable.

Two-Way ANOVA Test Before Log-Transformation for Hypothesis 1: (Parallel Design with Blocking Factor)

- **Model:** $total \sim Group_Label + facilitytype_nhc + Group_Label:facilitytype_nhc$

ANOVA Table:

Term	p-value
Group_Label	0.007
facilitytype_nhc	0.928
Group_Label:facilitytype_nhc	0.594

- **Interpretation:** The results showed a significant main effect of treatment ($p = 0.007$), indicating that CBVs in the treatment group referred and accompanied significantly more women compared to the control group. However, neither the main effect of facility type ($p = 0.928$) nor the interaction effect ($p = 0.594$) was statistically significant, suggesting that the treatment effect is consistent across facility types.

Tukey HSD Test

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	0.2329	0.8398	-2.0401	2.5059	False

The mean difference between groups was 0.23, with a p-adjusted value of 0.8398. The 95% confidence interval ranged from -2.04 to 2.51 , and the null hypothesis of equal means was not rejected. This indicates that there is no statistically significant difference in total referrals/accompaniments between the treatment and control groups when adjusting for multiple comparisons.

Assumption Check:

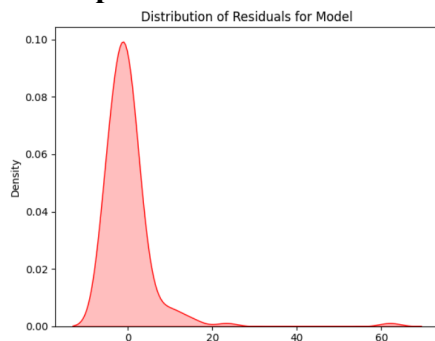


Figure 5. Density Plot

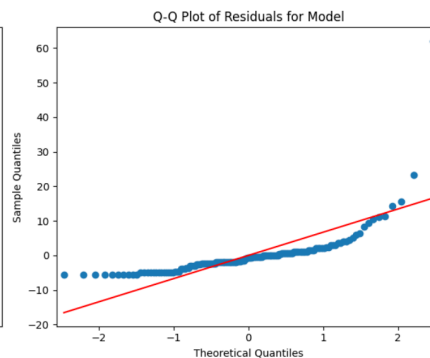


Figure 6. Q-Q Plot of Residuals

The density plot showed a right-skewed distribution and the Q-Q plot displayed substantial deviations from the theoretical quantiles. Additionally, the Shapiro–Wilk test

($W = 0.545$, $p < 0.001$) confirmed a statistically significant violation of the normality assumption.

Since the dependent variable (i.e. total) does not follow normal distribution, we must log transform it.

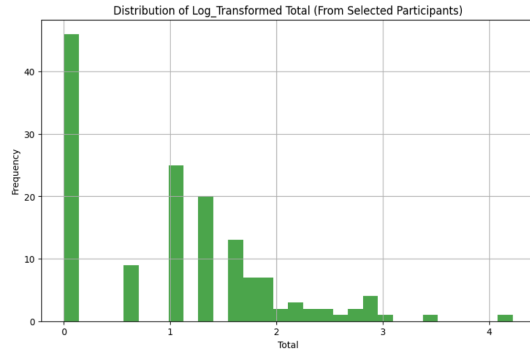


Figure 7. Histogram showing the distribution of log-transformed total referrals/accompaniments from selected participants. While still right-skewed, the transformation substantially compresses extreme values and improves the symmetry compared to the raw data.

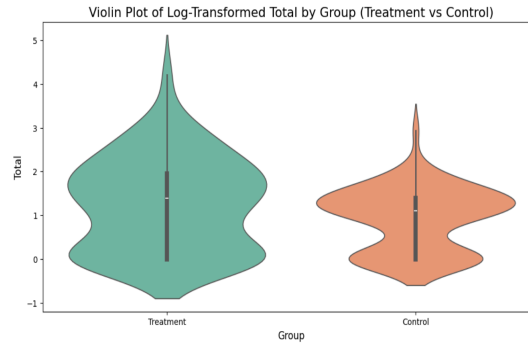


Figure 8. Violin plot of log-transformed total by treatment group (Treatment vs Control). The treatment group shows a slightly higher median and more spread in distribution, suggesting a potential effect of incentives on CBV referral activity after accounting for skewness.

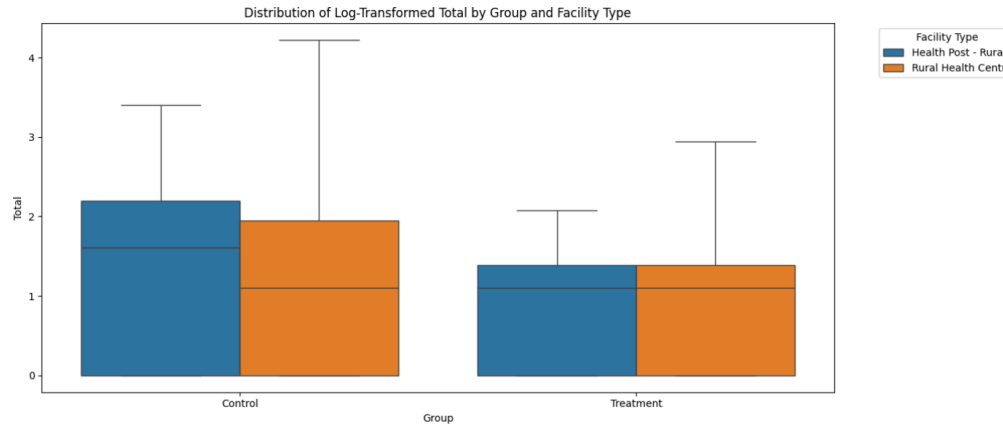


Figure 9. Boxplot of log-transformed total by treatment group and facility type. Both facility types show slightly higher central tendency in the control group, but variation remains across groups. This visualization supports exploring potential interaction effects using a two-way ANOVA.

Two-Way ANOVA Test After Log-Transformation for Hypothesis 1: (Parallel Design with Blocking Factor)

- **Model:** $total \sim Group_Label + facilitytype_nhc + Group_Label:facilitytype_nhc$

- **ANOVA Table:**

Term	p-value
Group_Label	0.033
facilitytype_nhc	0.551
Group_Label:facilitytype_nhc	0.207

- **Interpretation:** The results showed a significant main effect of treatment ($p = 0.033$), indicating that CBVs in the treatment group referred and accompanied significantly more women compared to the control group after log-transformation. However, neither the main effect of facility type ($p = 0.551$) nor the interaction effect between treatment and

facility type ($p = 0.207$) was statistically significant, suggesting that the treatment effect is consistent across different types of facilities.

- **Tukey HSD Test**

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	-0.0416	0.7849	-0.342	0.2589	False

The mean difference between groups was -0.0416 , with a p-adjusted value of 0.7849 . The 95% confidence interval ranged from -0.342 to 0.2589 , and the null hypothesis of equal means was not rejected. This indicates that there is no statistically significant difference in total referrals/accompaniments between the treatment and control groups when adjusting for multiple comparisons.

- **Assumption Check:**

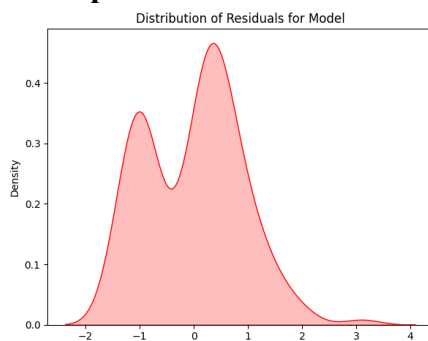


Figure 10. Density Plot

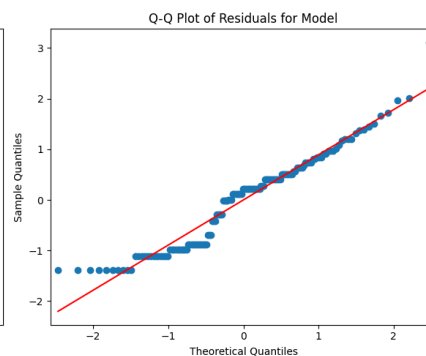


Figure 11. Q-Q Plot of Residuals

The density plot shows a bimodal but roughly symmetric distribution of residuals, while the Q-Q plot indicates that most points fall near the diagonal line, suggesting approximate normality with minor deviations. The Shapiro–Wilk test result ($W = 0.948$, $p < 0.001$) indicates a statistically significant departure from normality, despite the visual plots appearing relatively normal.

Parallel Design Setup with Continuous Covariates for Hypothesis 1

Objective: To determine whether providing financial incentives to community-based volunteers (CBVs) leads to an increase in total maternal health outreach activities (referrals and accompaniments), while controlling for potential variability due to gender representation in the NHC workforce.

Setup:

- **Continuous Covariate:**
 - `Femalenhc_ratio`: The proportion of members in the Neighborhood Health Committee (NHC) who are women. A higher proportion of women in NHCs may positively influence maternal health outreach, as female volunteers may be more approachable or relatable to expectant mothers. Including `femalenhc_ratio` as a covariate helps control for this potential source of variation and allows for a more accurate assessment of the independent effect of the incentive intervention.
- Blocking variable was not included in this setup to isolate covariate effects.

- **Randomization:** Participants are randomly assigned to one of two groups
- **Modified Hypothesis**
 - **Null Hypothesis (H_0):** After controlling for the proportion of members in the Neighborhood Health Committee (NHC) who are women, there is no significant difference in the total number of women referred and accompanied between the treatment and control groups.
 - **Alternative Hypothesis (H_a):** After controlling for the proportion of members in the Neighborhood Health Committee (NHC) who are women, the treatment group (CBVs receiving incentives) will refer and accompany a significantly higher number of women compared to the control group.

Type III ANOVA Before Log-Transformation for Hypothesis 1 (Parallel Design with Continuous Covariate)

- **Model:** $total \sim treatment_nhc + femalenhc_ratio + treatment_nhc:femalenhc_ratio$

ANOVA Table:

Term	p-value
Intercept	0.154
treatment_nhc	0.299
femalenhc_ratio	0.893
treatment_nhc:femalenhc_ratio	0.436

- **Interpretation:** The results indicate that the main effect of treatment was not statistically significant ($p = 0.299$), suggesting that CBVs who received monetary incentives did not refer or accompany significantly more women compared to those in the control group prior to log transformation. Additionally, the covariate *femalenhc_ratio*, which represents the proportion of female neighborhood health committee members, also did not show a significant effect ($p = 0.893$), indicating that variation in this proportion does not explain differences in referral counts. The interaction term between treatment and *femalenhc_ratio* was likewise non-significant ($p = 0.436$), suggesting that the effect of treatment does not depend on the proportion of female health committee members.

Assumption Check:

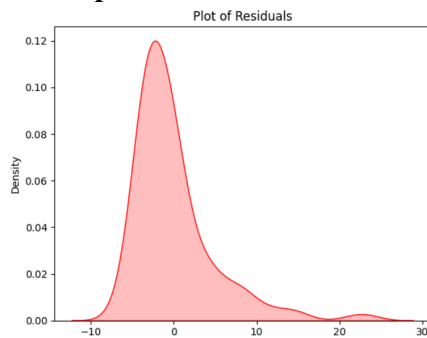


Figure 12. Density Plot

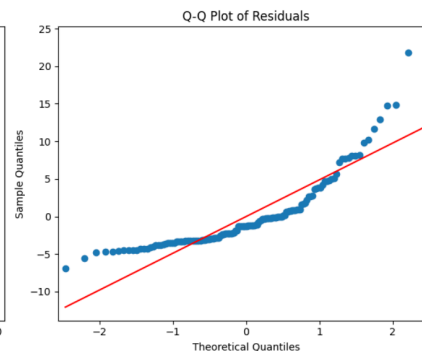


Figure 13. Q-Q Plot of Residuals

The density plot shows a right-skewed distribution of residuals, and the Q-Q plot shows notable deviations from the theoretical quantile line, particularly in the tails. Furthermore,

the Shapiro–Wilk test yielded a W statistic of 0.787 with a p-value < 0.001, indicating a statistically significant violation of the normality assumption. These results suggest that the residuals are not normally distributed.

Type III ANOVA After Log-Transformation for Hypothesis 1 (Parallel Design with Continuous Covariate)

- **Model:** $total \sim treatment_nhc + femalenhc_ratio + treatment_nhc:femalenhc_ratio$

- **ANOVA Table:**

Term	p-value
Intercept	0.003
treatment_nhc	0.643
femalenhc_ratio	0.568
treatment_nhc:femalenhc_ratio	0.664

- **Interpretation:** The results from the Type III ANOVA after log-transformation show that none of the model terms were statistically significant. The main effect of treatment (treatment_nhc) had a p-value of 0.643, indicating no significant difference in log-transformed referral counts between the treatment and control groups. Similarly, the covariate femalehc_ratio (p = 0.568) and its interaction with treatment (p = 0.664) were also not significant. These findings suggest that, after log-transforming the outcome variable to account for non-normality, there is no evidence that either treatment, the proportion of female NHC members, or their interaction had a statistically significant impact on the total number of referrals or accompaniments.

- **Assumption Check:**



Figure 14. Density Plot

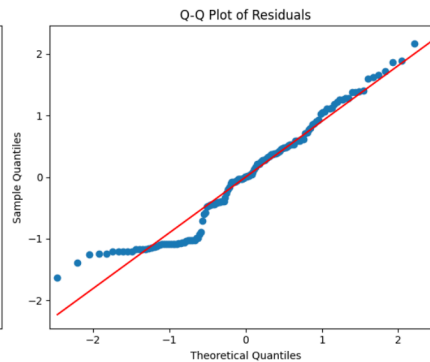


Figure 15. Q-Q Plot of Residuals

The density plot shows a bimodal and slightly left-skewed distribution, which departs from the ideal bell-shaped curve of normality. The Q-Q plot displays moderate deviations from the diagonal reference line, especially at the lower and upper tails, suggesting that the residuals are not perfectly normally distributed. Furthermore, the Shapiro–Wilk test yielded a W statistic of 0.955 with a p-value < 0.001, indicating a statistically significant violation of the normality assumption.

1.2 Hypothesis 2 FAC-Level From Merged DataFrame

Null Hypothesis (H_0): Facilities where CBVs receive monetary incentives (treatment group) will have a significantly higher number of women seeking Antenatal Care (ANC) within the first trimester compared to facilities where CBVs work voluntarily (control group).

Alternative Hypothesis (H_a): Facilities where CBVs receive monetary incentives (treatment group) will NOT have a significantly higher number of women seeking Antenatal Care (ANC) within the first trimester compared to facilities where CBVs work voluntarily (control group).

Subset Data Description:

Variable Name	Variable Type	Description	Why It Was Chosen
<code>treatment_fac</code>	Categorical (0 = control, 1 = treatment)	Whether the facility is in the treatment group	Independent variable
<code>anc_first_tri_num</code>	Continuous (Integer)	Number of women who received ANC in the first trimester at the facility	Dependent variable , reflects early maternal care engagement
<code>fid</code>	Categorical	Unique identifier for each facility or NHC	Used to group datasets across levels.
<code>Group_Label</code>	Categorical	Label for treatment/control group in some merged datasets (e.g., "Treatment")	For tracking treatment status in merged/reshaped views
<code>yr_month</code>	Categorical	Combined year-month timestamp	Used to align or aggregate temporal data accurately
<code>facilitytype_fac</code>	Categorical	Type of health facility	Used as a blocking factor to control for facility differences
<code>no_maternity_beds</code>	Continuous (Integer)	Number of maternity beds available at the facility	Covariate indicating delivery capacity
<code>distance_dmo</code>	Continuous (Float)	Distance from the facility to the District Medical Office	Covariate for geographic access and logistical challenges

Figure 16 illustrates the distribution of the dependent variable within the two groups (i.e. treatment vs. control) for the parallel design hypothesis 2 setup.

Parallel Design Setup with Blocking Factor for Hypothesis 2

Objective: To determine whether offering monetary incentives to community-based volunteers (CBVs) leads to a higher number of women seeking antenatal care (ANC) in the first trimester compared to a control group without incentives, while controlling for potential differences in service delivery due to facility type.

Setup:

- **Blocking Factor:** `facilitytype_fac` is chosen as the blocking factor once again since the outcome variable (i.e. `anc_first_tri_num`) could be dependent on the type of facility that the women are being placed in. We want to account for the variability within different types of facilities.

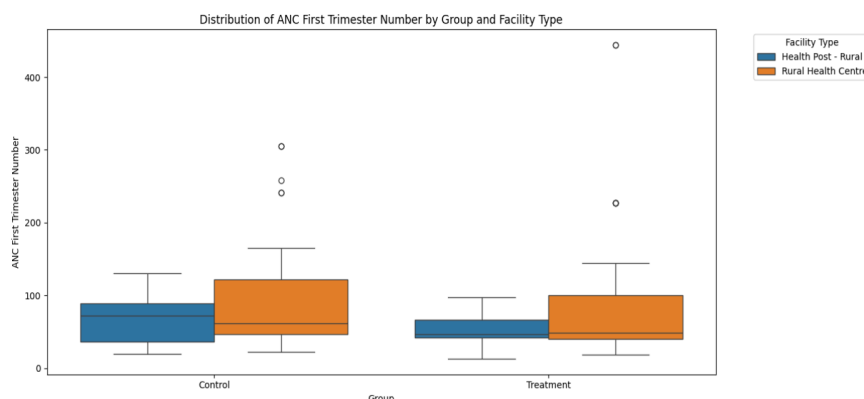
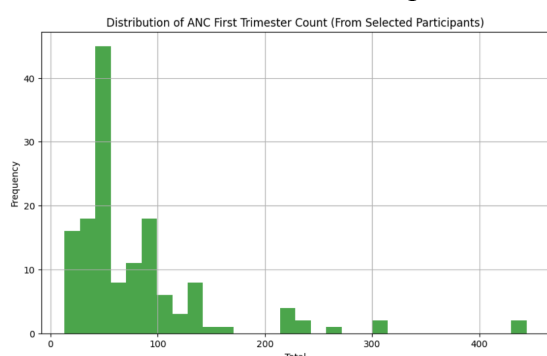


Figure 17. Box plot showing the distribution of first trimester ANC counts across treatment and control groups, segmented by facility type. The Rural Health Centre control group shows high variability and multiple extreme values, while treatment groups appear more stable. The plot highlights the importance of including `facilitytype_fac` as a blocking factor to account for structural differences in facility capacity and service delivery.

- **Randomization:** Participants are randomly assigned to one of two groups.



18. Histogram showing the distribution of ANC First Trimester Counts from selected participants. The data are right-skewed with a concentration of values between 30 and 100 and several outliers above 300, suggesting non-normality.

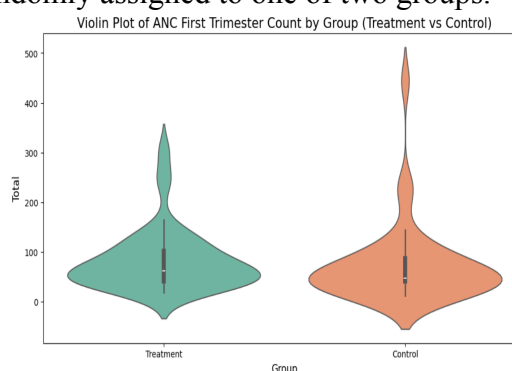


Figure 19. Violin plot comparing ANC First Trimester Counts between Treatment and Control groups. Both groups display similar medians, though the Control group shows more extreme high values. Further statistical analysis is needed to determine whether these differences are significant.

Two-Way ANOVA Test Before Log-Transformation for Hypothesis 2: (Parallel Design with Blocking Factor)

- **Model:** $anc_first_tri_num \sim Group_Label + facilitytype_fac + Group_Label: facilitytype_fac$

- **ANOVA Table:**

Term	p-value
Group_Label	0.126
facilitytype_fac	0.003
Group_Label:facilitytype_fac	0.325

- **Interpretation:** The main effect of the treatment group (`Group_Label`) was not statistically significant ($p = 0.126$), suggesting that there is no strong evidence that providing financial incentives to CBVs significantly increased ANC first trimester counts across all facility types. However, the main effect of facility type (`facilitytype_fac`) was statistically significant ($p = 0.003$), indicating that differences in ANC counts are

significantly associated with the type of facility. The interaction between treatment group and facility type ($p = 0.325$) was not significant, suggesting that the effect of treatment is consistent across different types of facilities.

- **Tukey HSD Test**

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	17.0959	0.1364	-5.4651	39.6569	False

The test found no statistically significant difference between the groups ($p\text{-adj} = 0.1364$). The 95% confidence interval for the mean difference ranged from -5.47 to 39.66, and the null hypothesis was not rejected. This confirms the earlier ANOVA finding that the treatment intervention did not significantly impact ANC uptake.

- **Assumption Check:**

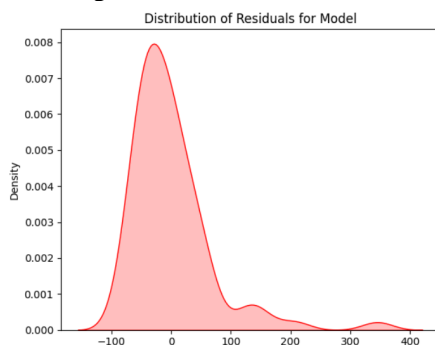


Figure 20. Density Plot

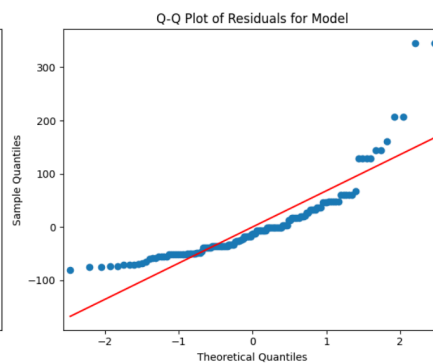


Figure 21. Q-Q Plot of Residuals

Both the residual density plot and the Q-Q plot indicated non-normality, with visible skewness and outliers. This was confirmed by the Shapiro–Wilk test ($W = 0.77$, $p < 0.001$), suggesting a significant deviation from normality.

Since the dependent variable (i.e. total) does not follow normal distribution, we must log transform it.

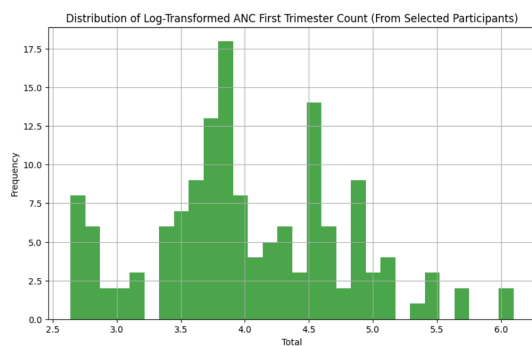


Figure 22. Histogram of Log-Transformed ANC First Trimester Count: The distribution appears approximately symmetric, with the majority of values concentrated between 3.5 and 4.5. Compared to the untransformed version, this transformation reduces right skewness and potential outliers, helping the data better approximate normality.

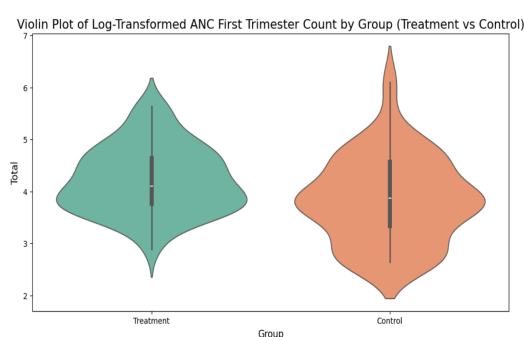


Figure 23. Violin plot compares the distribution of log-transformed ANC counts between the treatment and control groups. Both groups show similar spread and central tendencies, with overlapping interquartile ranges. The treatment group appears to have a slightly higher median, but the overall distribution shape is quite similar.

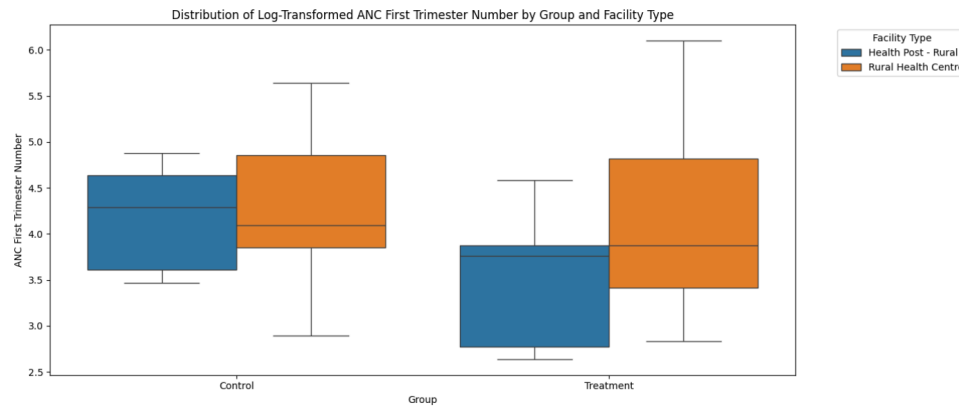


Figure 24. Boxplot of log-transformed total by treatment group and facility type. Although some differences exist in medians and spread between facility types, the treatment and control groups within each facility type are quite comparable. This visualization complements the ANOVA finding that facility type plays a significant role in ANC counts, while treatment group and the interaction effect do not.

Two-Way ANOVA Test After Log-Transformation for Hypothesis 2: (Parallel Design with Blocking Factor)

- **Model:** $anc_first_tri_num \sim Group_Label + facilitytype_fac + Group_Label:facilitytype_fac$

- **ANOVA Table:**

Term	p-value
Group_Label	0.125
facilitytype_nhc	0.003
Group_Label:facilitytype_nhc	0.325

- **Interpretation:** The results show that the main effect of group (treatment vs control) is not statistically significant ($p = 0.125$), suggesting that offering financial incentives to CBVs did not lead to a significantly higher number of women seeking ANC services in the first trimester after accounting for facility type. However, the main effect of facility type is statistically significant ($p = 0.003$), indicating that the type of health facility (e.g., Health Post – Rural vs Rural Health Centre) has a significant influence on ANC first trimester counts. The interaction effect between group and facility type is not significant ($p = 0.325$), implying that the impact of treatment is consistent across facility types.

- **Tukey HSD Test**

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	0.3867	0.0018	0.1462	0.6271	True

The Tukey HSD test revealed a statistically significant difference between the treatment and control groups ($mean\ difference = 0.3867$, $p\text{-adjusted} = 0.0018$). The 95% confidence interval for the difference ranged from 0.1462 to 0.6271, and the null hypothesis of equal means was rejected. This indicates that facilities where CBVs received monetary incentives had significantly higher log-transformed ANC first trimester counts compared to the control group, even after adjusting for multiple comparisons.

- **Assumption Check:**

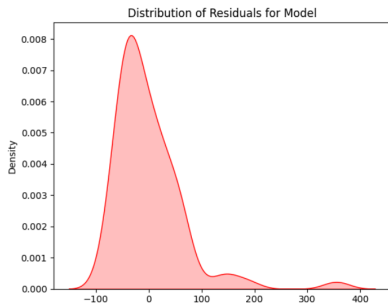


Figure 25. Density Plot

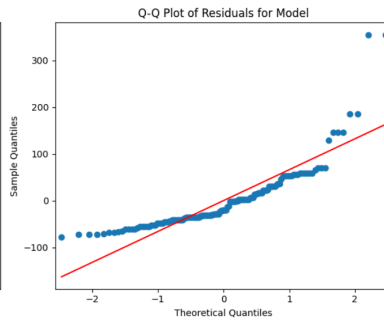


Figure 26. Q-Q Plot of Residuals

The density plot shows a strong right-skewed distribution of residuals, and the Q-Q plot reveals substantial deviations from the diagonal line, particularly in the tails. The Shapiro–Wilk test result ($W = 0.770$, $p < 0.001$) confirms a statistically significant violation of the normality assumption.

- **Justification For Choosing Same Blocking Factor For Both Hypothesis 1 & 2:** The decision to use the same blocking factor for both hypotheses was made since facility type was a feature in both the NHC and FAC level datasets. As a result, it is critical to account for such a significant feature as a blocking factor in both hypotheses.

Parallel Design Setup with Continuous Covariates for Hypothesis 2

Objective: To determine whether providing monetary incentives to CBVs leads to a higher number of antenatal care (ANC) visits in the first trimester compared to the control group, while controlling for facility-level characteristics such as the number of maternity beds and distance to the District Medical Office (DMO), which may affect service capacity and accessibility.

Setup:

- **Continuous Covariate:**
 - **no_maternity_beds:** Number of maternity beds at the facility. The availability of maternity beds could influence how many women can be accommodated at a given time, affecting the facility's ability to provide ANC services. A higher number of maternity beds might lead to a higher capacity for ANC services, which could influence the number of women seeking ANC and, consequently, the number of women referred by CBVs.
 - **distance_dmo:** Distance of the facility from the District Medical Office. The distance to the District Medical Office (DMO) could affect the accessibility of healthcare services, including ANC. If a facility is farther from the DMO, it might face challenges in providing ANC services or facilitating referrals, which could influence the outcome.
- Blocking variable was not included in this setup to isolate covariate effects.
- **Randomization:** Participants are randomly assigned to one of two groups
- **Modified Hypothesis**
 - **Null Hypothesis (H_0):** After controlling for the number of maternity beds and distance to the District Medical Office (DMO), there is no significant difference

in the number of women receiving ANC in the first trimester between treatment and control facilities

- **Alternative Hypothesis (H_a):** After controlling for the number of maternity beds and distance to the District Medical Office (DMO), there is a significant difference in the number of women receiving ANC in the first trimester between treatment and control facilities

Type III ANOVA Before Log-Transformation for Hypothesis 2 (Parallel Design with Continuous Covariates)

- **Model:** $anc_first_tri_num \sim treatment_fac + no_maternity_beds + distance_dmo + treatment_fac:no_maternity_beds + treatment_fac:distance_dmo + no_maternity_beds:distance_dmo$

● Anova Table:

Term	p-value
Intercept	0.035
treatment_fac	0.460
no_maternity_beds	0.219
treatment_fac:no_maternity_beds	0.089
distance_dmo	0.409
treatment_fac:distance_dmo	0.486
no_maternity_beds:distance_dmo	0.990

- **Interpretation:** The results indicate that the main effect of treatment (treatment_fac) was not statistically significant ($p = 0.460$), suggesting that monetary incentives for CBVs did not significantly impact the number of women seeking ANC in the first trimester when controlling for covariates. Likewise, neither no_maternity_beds ($p = 0.219$) nor distance_dmo ($p = 0.409$) had a significant effect individually. All interaction terms, treatment_fac:no_maternity_beds ($p = 0.089$), treatment_fac:distance_dmo ($p = 0.486$), and no_maternity_beds:distance_dmo ($p = 0.990$), were also not significant, indicating no meaningful moderating effect among these continuous covariates. Overall, none of the predictors in the model explained a significant portion of the variation in ANC first trimester counts.

● Assumption Check:

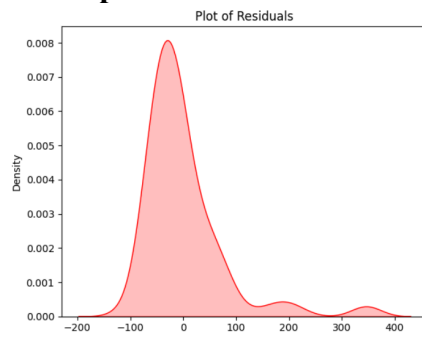


Figure 27. Density Plot

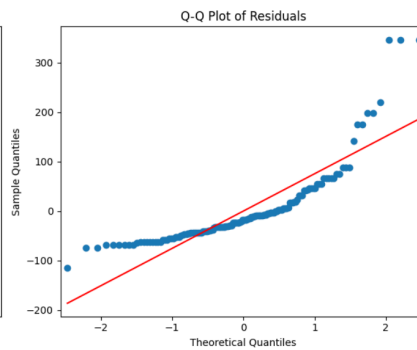


Figure 28. Q-Q Plot of Residuals

The density plot displays a right-skewed distribution of residuals, and the Q-Q plot shows considerable deviations from the theoretical quantile line, particularly in the tails, indicating non-normality. This visual evidence is supported by the Shapiro–Wilk test result ($W = 0.741$, $p < 0.001$), which confirms a statistically significant departure from normality.

Type III ANOVA After Log-Transformation for Hypothesis 2 (Parallel Design with Continuous Covariates)

- **Model:** $anc_first_tri_num \sim treatment_fac + no_maternity_beds + distance_dmo + treatment_fac:no_maternity_beds + treatment_fac:distance_dmo + no_maternity_beds:distance_dmo$

- **Anova Table:**

Term	p-value
Intercept	0.000
treatment_fac	0.046
no_maternity_beds	0.057
treatment_fac:no_maternity_beds	0.039
distance_dmo	0.894
treatment_fac:distance_dmo	0.697
no_maternity_beds:distance_dmo	0.484

- **Interpretation:** The results indicate that the main effect of treatment ($C(treatment_fac)$) was statistically significant ($p = 0.0458$), suggesting that monetary incentives for CBVs had a significant impact on the number of women seeking ANC in the first trimester after adjusting for covariates. However, neither $no_maternity_beds$ ($p = 0.0568$) nor $distance_dmo$ ($p = 0.894$) were significant individual predictors. All the interaction terms were not significant except $C(treatment_fac):no_maternity_beds$ was statistically significant ($p = 0.0392$), indicating that the effect of treatment varied depending on the number of maternity beds.

- **Assumption Check:**

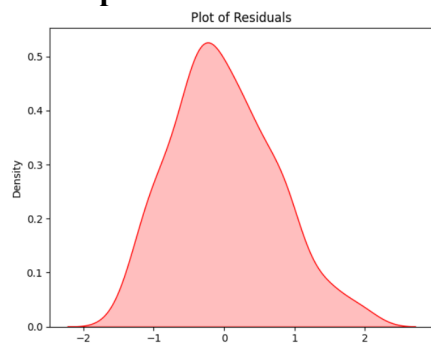


Figure 29. Density Plot

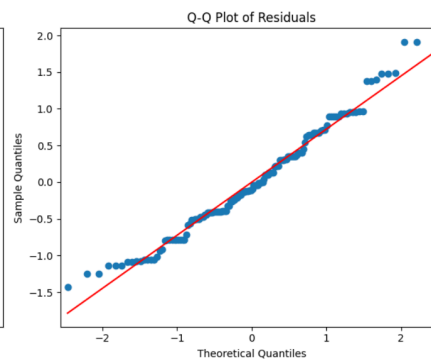


Figure 30. Q-Q Plot of Residuals

The density plot shows a nearly symmetric and bell-shaped distribution of residuals, indicating that the residuals are approximately normally distributed. The Q–Q plot further

supports this, as the residual points align closely along the reference line, with only minor deviations at the tails. The Shapiro–Wilk test yields a test statistic of $W = 0.977$ and a $p\text{-value} = 0.013$, which is statistically significant at the 5% level. This suggests a slight deviation from perfect normality.

3. Matched Pair Design

Our matched pair design used "number of maternity beds" and "population in catchment area" as the matching criteria to form matched pairs of facilities, and all the facilities remain in their assigned group throughout the entire observation period. To achieve medium effect, a total of 146 participants should be selected from the merged data to form 73 pairs. However, there is only a sum of 98 facilities which leads to 49 pairs. Random participants will be selected from the pairs and the other one in the pair will be assigned to another group such that:

- Group 1 (Treatment - Incentives, 49 participants)
- Group 2 (Control - No Incentives, 49 participants)

We selected the matched pair design because it helped control for baseline facility differences in size and population served, which improved statistical power and strengthened the ability to attribute differences in numbers of women who sought first antenatal care, rather than to structural differences between facilities.

We used different hypotheses for Hypothesis 1 and Hypothesis 2. Although both hypotheses evaluate the impact of monetary incentives on women's decisions to take ANC services, they are tested at different levels of analysis and capture distinct behavioral outcomes. Similar to the ones in parallel design, hypothesis 1 focuses on the outcome of the facility (incentives to CBVs ultimately increase the number of women arriving for ANC in the first trimester). This outcome depends on facility accessibility, readiness, and women's follow-through. Hypothesis 2 focuses on the CBV behaviour (whether incentives lead to more referrals or accompaniment of women to health facilities). This reflects the output of the CBV directly and is measured using the total number of women referred.

2.1 Hypothesis 1 Facility-Level From Merged DataFrame

Null Hypothesis (H_0): The intervention/treatment (incentives for CBVs) will lead to an increase in the total number of women who sought ANC at that facility over the entire period.

Alternative Hypothesis (H_a): The intervention/treatment (incentives for CBVs) will NOT lead to an increase in the total number of women who sought ANC at that facility over the entire period .

Subset Data Description:

Variable Name	Variable Type	Description	Why It Was Chosen
treatment	Categorical (0 = control, 1 = treatment)	Indicates whether the facility received financial incentives	Independent variable

anc_first_tri_num	Continuous (Integer)	Total women who sought ANC at that facility over the entire period	Dependent variable
fid	Categorical	Unique identifier for each facility or NHC	Used to group datasets across levels.
d	Categorical	District	Decide which district the facility belongs to for region identification
Group_Label	Categorical	Label for treatment/control group in some merged datasets (e.g., “Treatment”)	For tracking treatment status in merged/reshaped views
region	Categorical	Location of the facility	Used as a blocking variable to account for the variability within different locations of facilities.
new_pair	Categorical	New pair numbers after accounting for blocking factor	Used for matched pair design when accounting for blocking factor
distance_dmo	Continuous	Distance to district medical office	Covariate for geographic access and logistical challenges.

Figure 31 shows the Control group has a slightly wider and more right-skewed distribution, suggesting a greater variability and potential for higher values. The median total appears somewhat higher in the Treatment group, though the overlap in distributions suggests that the difference may be modest.

Matched Pair Design Setup with Blocking Factor for Hypothesis 1

Objective: To determine whether offering financial incentives to community-based volunteers (CBVs) leads to a higher number of women who sought ANC at that facility over the entire period to a control group with no incentives, while controlling for potential regional effects using “region” as a blocking factor.

Setup:

- **Blocking Factor:** “region” is created as the blocking factor since the outcome variable (i.e. total women who sought for ANC) could be dependent on the location of facility that the womens are being placed in. We want to account for the variability within different regions of facilities.

Figure 32 illustrates that, in the north region, the control group exhibits higher median referral counts and a wider spread compared to the treatment group; while in south region, the control group exhibits lower median referral counts and a wider spread compared to the treatment group. The South region generally shows higher variability, while the North region shows more extreme outliers. This suggests that referral outcomes may differ depending on both treatment condition and regions, justifying the inclusion of regions as a blocking factor.

- **Randomization:** Participants in pairs are randomly assigned to one of two groups.

[Figure 33](#) shows the data are right-skewed with a Concentration of values between 0 and 160 and a few extreme outliers above 200, indicating non-normality in CBV referral activity.

[Figure 34](#) shows the treatment group has a slightly higher medium effect while the control group shows greater variability and heavier tails, suggesting that there are no significant differences for dependent variables on whether receiving incentives or not.

Two-Way ANOVA Test for Hypothesis 1 (Matched Pair Design with Blocking Factor)

- **Model:** $anc_first_tri_num \sim Group_Label + region + Group_Label:region$

- **ANOVA Table:**

Term	p-value
Group_Label	0.887
region	0.507
Group_Label:region	0.824

- **Interpretation:** The results showed no significant main effect of treatment ($p = 0.887$), indicating that CBVs in the treatment group showed no significantly more women seeking for ANC compared to the control group. Neither the main effect of region ($p = 0.507$) nor the interaction effect ($p = 0.824$) was statistically significant, suggesting that the treatment effect is consistent across regions.

- **Tukey HSD Test**

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	2.0426	0.8863	-26.2587	30.3438	False

The mean difference between groups was 2.04, with a p-adjusted value of 0.8863. The 95% confidence interval ranged from -26.26 to 30.34, and the null hypothesis of equal means was not rejected. This indicates that there is no statistically significant difference in total women who sought for ANC between the treatment and control groups when adjusting for multiple comparisons.

- **Assumption Check:**

[Figure 35](#) the density plot showed a right-skewed distribution and [figure 36](#) the Q-Q plot displayed substantial deviations from the theoretical quantiles. Additionally, the Shapiro–Wilk test ($W = 0.758$, $p < 0.001$) confirmed a statistically significant violation of the normality assumption.

Matched Pair Design Setup with Continuous Covariates for Hypothesis 1

Objective: To determine whether providing financial incentives to community-based volunteers (CBVs) leads to an increase in total women who sought for ANC, while controlling for potential variability due to distance to the district office.

Setup:

- **Continuous Covariate:**

- *distance_dmo*: The distance of the facility to the district medical office. A longer distance of the facility to the district medical office may positively influence maternal health outreach, as women may seek for ANC when they are closer to the district medical office. Including *distance_dmo* as a covariate helps control for this potential source of variation and allows for a more accurate assessment of the independent effect of the incentive intervention.

- Blocking variable was not included in this setup to isolate covariate effects.

- **Randomization:** Participants in pairs are randomly assigned to one of two groups

- **Modified Hypothesis**

- **Null Hypothesis (H_0):** After controlling for the distance to the district medical office, there is no significant difference in the total number of women who sought for ANC between the treatment and control groups.

- **Alternative Hypothesis (H_a):** After controlling for the distance to the district medical office, the treatment group (CBVs receiving incentives) will have more women seeking for ANC compared to the control group.

ANOVA for Hypothesis 1 (Matched Design with Continuous Covariate)

- **Model:** $anc_first_tri_num \sim C(Group_Label) + distance_dmo$

- **ANOVA Table:**

Term	p-value
Intercept	<0.001
C(Group_Label)	0.701
distance_dmo	0.258

- **Interpretation:** The results indicate that the main effect of treatment was not statistically significant ($p = 0.701$), suggesting that CBVs who received monetary incentives did not increase the numbers of women who sought for ANC compared to those in the control group. Additionally, the covariate *distance_dmo*, which represents the distance to the medical office, also did not show a significant effect ($p = 0.258$),

indicating that variation does not explain differences in the number of women who sought for ANC.

- **Assumption Check:**

Figure 37 The density plot shows a right-skewed distribution of residuals, and **figure 38** the Q-Q plot shows notable deviations from the theoretical quantile line, particularly in the tails. Furthermore, the Shapiro–Wilk test yielded a W statistic of 0.776 with a p-value < 0.001, indicating a statistically significant violation of the normality assumption. These results suggest that the residuals are not normally distributed.

2.2 Hypothesis 2 NHC-Level From Merged DataFrame

Null Hypothesis (H_0): The intervention/treatment (incentives for CBVs) will lead to an increase in the total number of women referred.

Alternative Hypothesis (H_a): The intervention/treatment (incentives for CBVs) will NOT lead to an increase in the total number of women referred.

Subset Data Description:

Variable Name	Variable Type	Description	Why It Was Chosen
treatment_nhc	Categorical (0 = control, 1 = treatment)	Indicates whether the facility received financial incentives	Independent variable
referred	Continuous (Integer)	Total number of women referred	Dependent variable
fid	Categorical	Unique identifier for each facility or NHC	Used to group datasets across levels.
d	Categorical	District	Decide which district the facility belongs to for region identification
Group_Label	Categorical	Label for treatment/control group in some merged datasets (e.g., “Treatment”)	For tracking treatment status in merged/reshaped views
region	Categorical	Location of the facility	Used as a blocking variable to account for the variability within different locations of facilities.
pop_total	Continuous (Integer)	Total population in nhc catchment zone	Provide population information for the districts
pair_id	Continuous	Pair number after accounting for blocking factor	Used for matched pair design when accounting for blocking factor
femalesmag_ratio	Continuous	Proportion female in SMAG	Covariate for capturing potential gender influence on referral

Figure 39 shows the Treatment group has a slightly wider and more right-skewed distribution, suggesting greater variability and potential for higher values. The median total appears similar in the Treatment group, which suggests that the difference may be modest.

Matched Pair Design Setup with Blocking Factor for Hypothesis 2

Objective: To determine whether offering financial incentives to community-based volunteers (CBVs) leads to a higher number of women referred to a control group with no incentives, while controlling for potential regional effects using “region” as a blocking factor.

Setup:

- **Blocking Factor:** “region” is created as the blocking factor since the outcome variable (i.e. number of women referred) could be dependent on the location of the facility that the women are being placed in. We want to account for the variability within different regions of facilities.

Figure 40 the boxplot illustrates that, in the north region, both groups have similar numbers of cases referred while the treatment group has a wider variability; while in the south region, the control group exhibits higher median referral counts and a wider spread compared to the treatment group. The South region generally shows higher variability and more extreme outliers. This suggests that referral outcomes may differ depending on both treatment conditions and regions, justifying the inclusion of regions as a blocking factor.

- **Randomization:** Participants in pairs are randomly assigned to one of two groups

Figure 41 shows the data are right-skewed with a Concentration of values between 0 and 5 and a few extreme outliers on 7, indicating non-normality in numbers of women referred.

Figure 42 shows the treatment group has a slightly lower medium effect and greater variability with heavier tails, suggesting that there is a difference for the number of cases referred to whether receiving incentives or not.

Two-Way ANOVA Test for Hypothesis 2 (Matched Pair Design with Blocking Factor)

- **Model:** $referred \sim Group_Label + region + Group_Label:region$
- **ANOVA Table:**

Term	p-value
Group_Label	0.374
region	<0.001
Group_Label:region	0.030

- **Interpretation:** The results showed no significant main effect of treatment ($p = 0.374$), indicating that the treatment group showed no significantly more women referred

compared to the control group. The region ($p < 0.001$) and interaction term ($p = 0.030$) are significant, suggesting that the treatment effect is imbalanced across regions.

- **Tukey HSD Test**

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
Control	Treatment	0.2055	0.4030	-0.2787	0.6897	False

The mean difference between groups was 0.21, with a p-adjusted value of 0.40. The 95% confidence interval ranged from -0.28 to 0.69 , and the null hypothesis of equal means was not rejected. This indicates that there is no statistically significant difference in total cases referred between the treatment and control groups when adjusting for multiple comparisons.

- **Assumption Check:**

[Figure 43](#) the density plot showed a right-skewed distribution and [figure 44](#) the Q-Q plot displayed substantial deviations from the theoretical quantiles. Additionally, the Shapiro–Wilk test ($W = 0.86$, $p < 0.001$) confirmed a statistically significant violation of the normality assumption.

Matched Pair Design Setup with Continuous Covariates for Hypothesis 2

Objective: To determine whether providing financial incentives to community-based volunteers (CBVs) leads to an increase in the number of cases referred, while controlling for potential variability due to the proportion of women in SMAG.

Setup:

- **Continuous Covariate:**

- **femalesmag_ratio:** The proportion of women in the Safe Motherhood Action Group. The more women in the SMAG may positively influence cases referred, as female volunteers may be more approachable or relatable to expectant mothers. Including **femalesmag_ratio** as a covariate helps control for this potential source of variation and allows for a more accurate assessment of the independent effect of the incentive intervention.

- Blocking variable was not included in this setup to isolate covariate effects.

- **Randomization:** Participants in pairs are randomly assigned to one of two groups

- **Modified Hypothesis**

- **Null Hypothesis (H_0):** After controlling for the proportion of women in the SMAG, there is no significant difference in the total number of cases referred between the treatment and control groups.

- **Alternative Hypothesis (H_a):** After controlling for the proportion of women in the SMAG, the treatment group will have more cases referred to compared to the control group.

ANOVA for Hypothesis 2 (Matched Design with Continuous Covariate)

- **Model:** $referred \sim treatment_nhc + femalesmag_ratio + treatment_nhc:femalesmag_ratio$
- **ANOVA Table:**

Term	p-value
Intercept	0.2044
treatment_nhc	0.0066
femalesmag_ratio	0.0058
treatment_nhc:femalesmag_ratio	0.0143

- **Interpretation:** The results indicate that the main effect of treatment was statistically significant ($p = 0.0066$), suggesting that CBVs who received monetary incentives increased the numbers of cases referred compared to those in the control group. Additionally, the covariate *femalesmag_ratio*, which represents the proportion of women in the SMAG, also showed a significant effect ($p = 0.0058$), indicating that variation does explain differences in the number of cases referred.

- **Assumption Check:**

[Figure 45](#) the density plot shows a right-skewed distribution of residuals, and [figure 46](#) the Q-Q plot shows notable deviations from the theoretical quantile line, particularly in the tails. Furthermore, the Shapiro–Wilk test yielded a W statistic of 0.779 with a p-value < 0.001 , indicating a statistically significant violation of the normality assumption. These results suggest that the residuals are not normally distributed.

4. Conclusion

In conclusion, the Parallel Design analyses showed mixed yet notable patterns: at the NHC level (Hypothesis 1), treatment groups generally referred and accompanied more women to ANC than controls (p-values of 0.007 before log-transformation and 0.033 after), but subsequent Tukey HSD tests did not confirm a statistically significant difference once multiple comparisons were considered. Meanwhile, at the facility level (Hypothesis 2), the untransformed analysis initially found no significant main effect for incentives ($p=0.126$), but log-transformation and a Tukey HSD test revealed that treatment facilities did have higher first-trimester ANC counts ($p\text{-adjusted}=0.0018$). By contrast, the Matched Pair Design results indicated no significant increase in total ANC-seeking behavior (facility-level Hypothesis 1) under incentives alone, especially once assumptions of normality were tested. However, for the NHC-level outcome (Hypothesis 2), adding the proportion of women in the Safe Motherhood Action Groups (*femalesmag_ratio*) as a covariate unveiled a significant main effect for incentives ($p=0.0066$),

suggesting that monetary payments to CBVs may indeed boost referrals when accounting for volunteer composition. Taken together, these findings underscore that incentives can improve some maternal health outcomes under certain conditions—particularly when log-transformations address skewed data and relevant covariates (i.e. facility type or volunteer gender ratio) are considered but their impact is not thoroughly robust across all settings or analytic approaches.

Work Cited

[1] *Intranet | Partnership for Economic Policy*. (2016). Pep-Net.org.
<https://portal.pep-net.org/public/project/20379>

[2] Websites used as references for coding certain parts of this project are directly cited in the Colab notebook, above the sections in which they were used.

Appendix A: Some Figures from Parallel and Matched Pairs Design

A.1: Parallel Design Figures

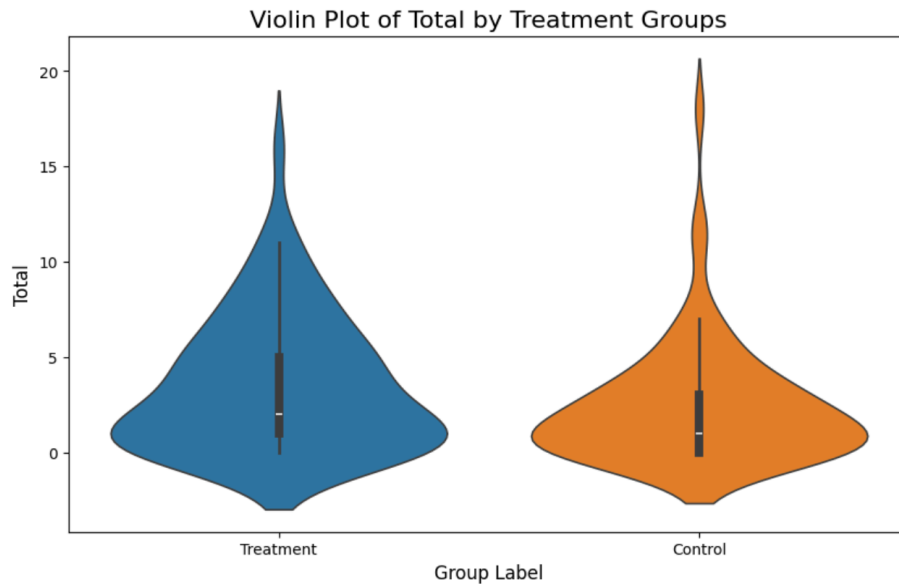


Figure 1. Violin plot comparing the distribution of total outreach activities between CBVs in the treatment and control groups. The Treatment group has a slightly wider and more right-skewed distribution, suggesting a greater variability and potential for higher outreach values. The median total appears somewhat higher in the Treatment group, though the overlap in distributions suggests that the difference may be modest.

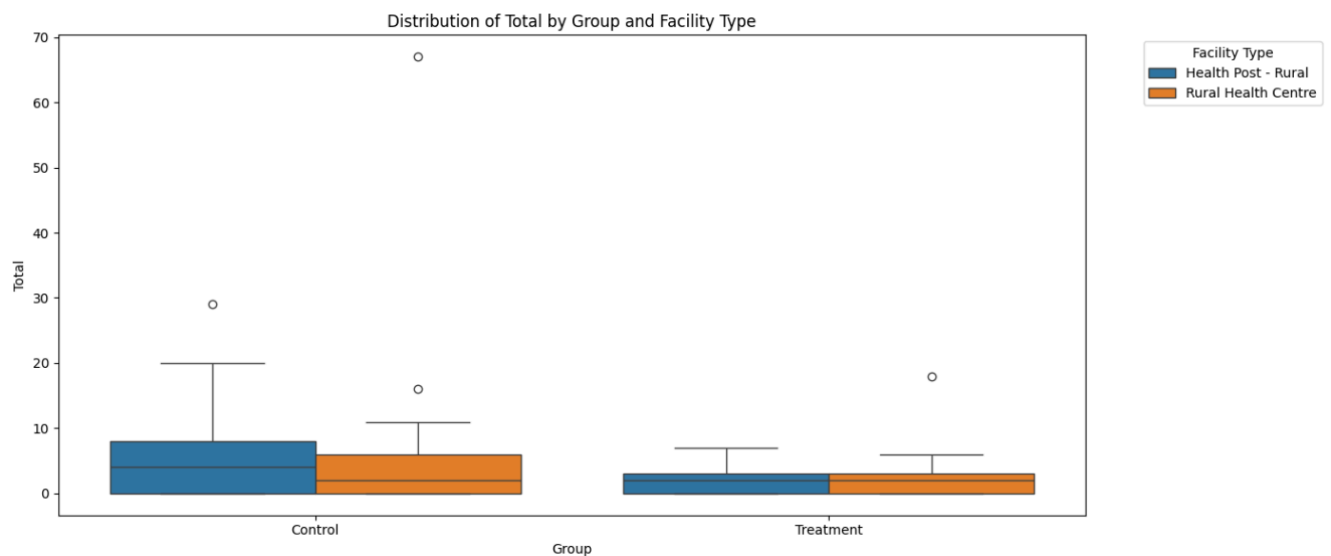


Figure 2. Boxplot showing the distribution of total referrals/accompaniments by treatment group and facility type. The plot illustrates that, across both facility types, the control group exhibits higher median referral counts and a wider spread compared to the treatment group. Health Posts generally show higher variability and more extreme outliers than Rural Health Centres. This suggests that referral outcomes may differ depending on both treatment condition and facility type, justifying the inclusion of facility type as a blocking factor.

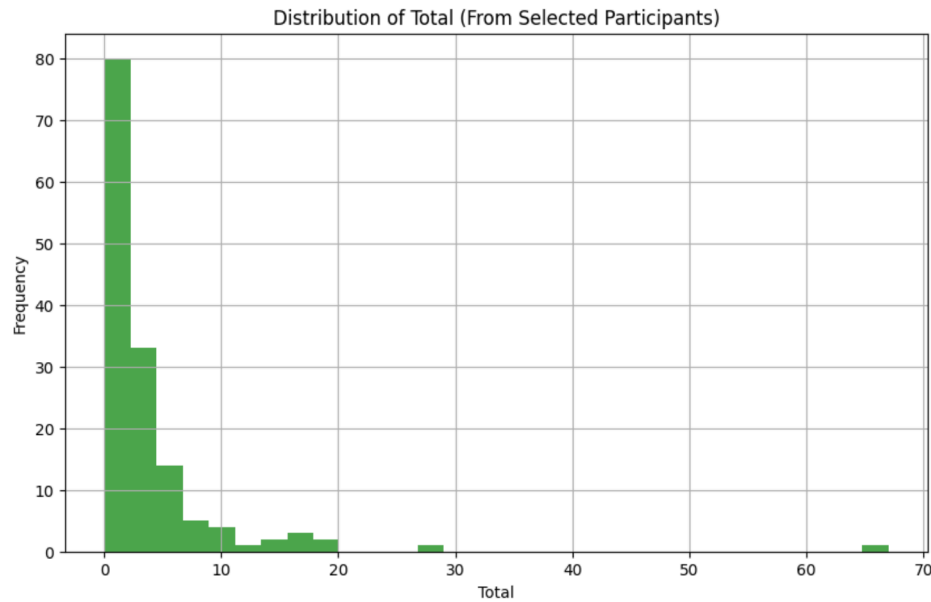


Figure 3. Histogram showing the distribution of total referrals/accompaniments from selected participants. The data are right-skewed with a concentration of values between 0 and 5 and a few extreme outliers above 60, indicating non-normality in CBV referral activity.

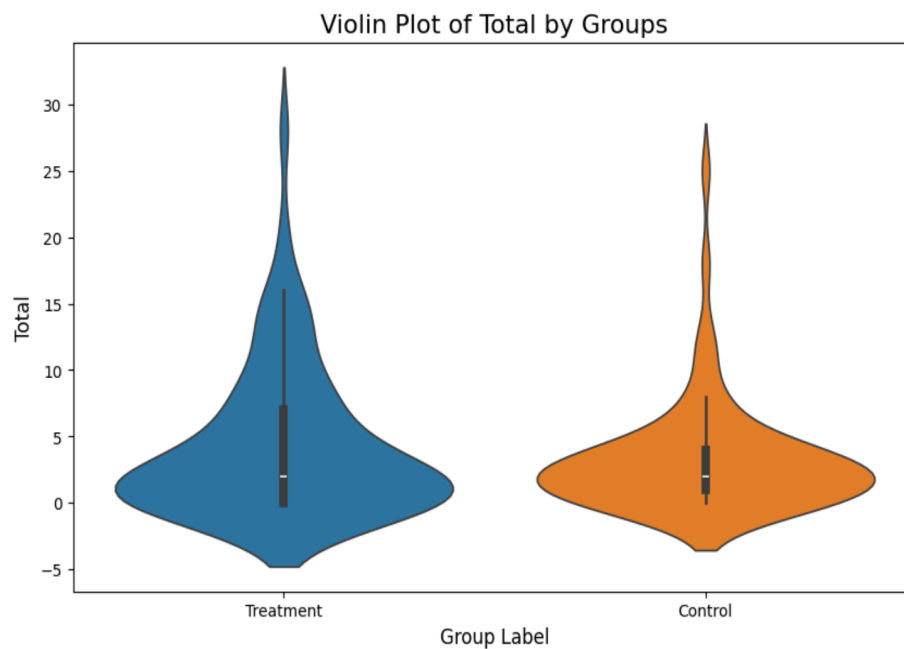


Figure 4. Violin plot showing the distribution of total referrals/accompaniments by treatment group (Treatment vs Control). Both groups exhibit similar central tendencies, but the treatment shows greater variability and heavier tails, suggesting that incentives may lead to increased referral activity for a subset of CBVs

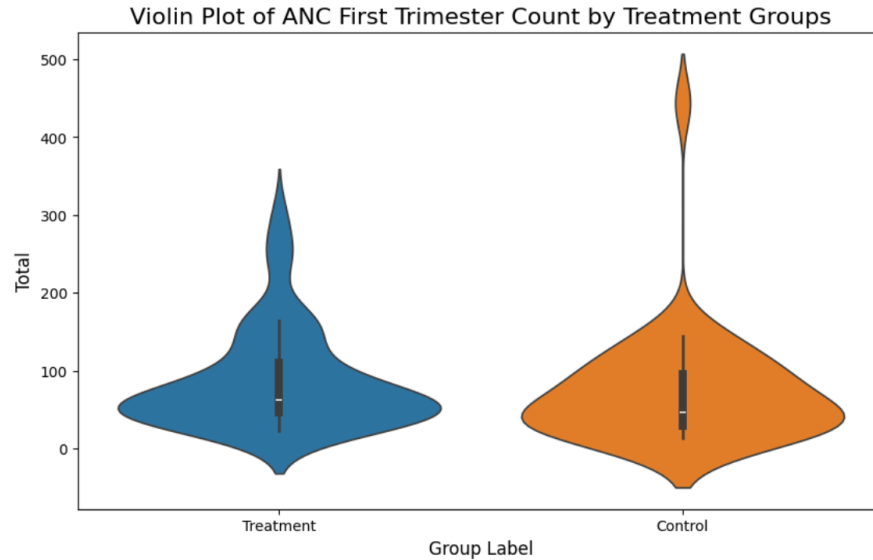


Figure 16. Violin plot comparing the distribution of ANC first trimester counts between Treatment and Control groups. Both groups show a wide range of values, though the Control group includes more extreme outliers. The median values are similar across groups, but the density shapes suggest greater consistency in the Treatment group. This visualization motivates further statistical analysis to assess whether treatment had a significant effect on early antenatal care uptake.

A.2: Matched Pairs Design Features

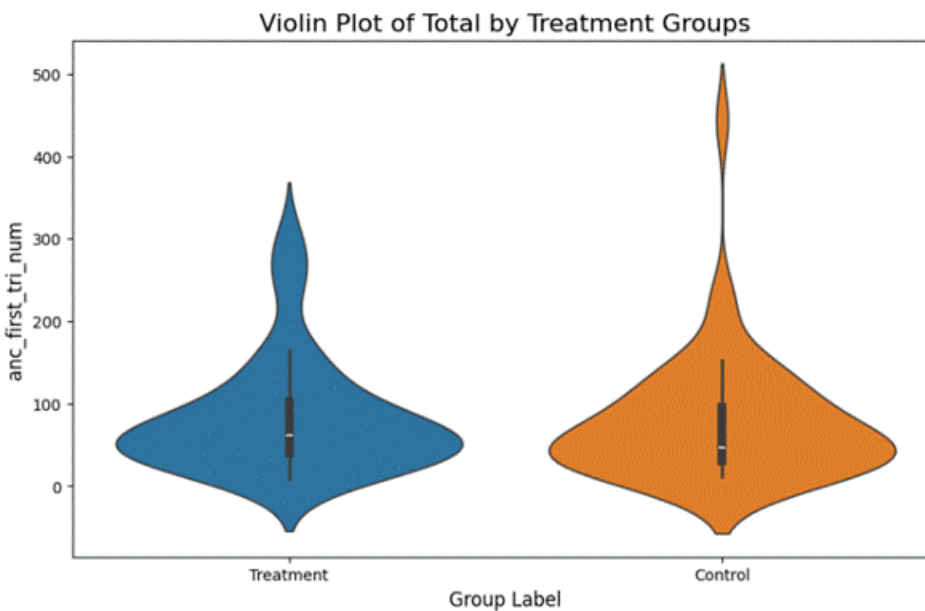


Figure 31. Violin plot comparing the distribution of total women who sought ANC at that facility over the entire period in the treatment and control groups

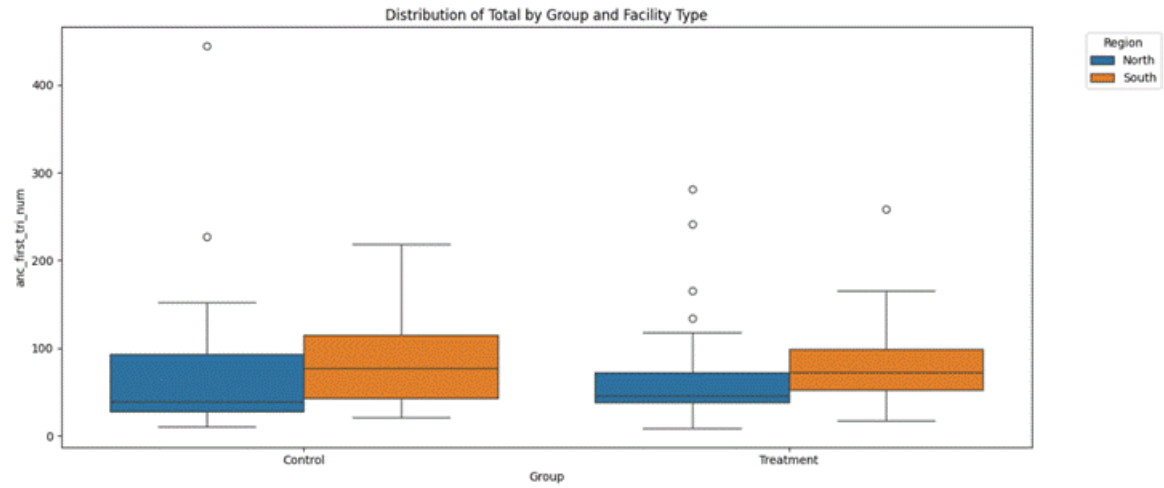


Figure 32. Boxplot showing the distribution of total women who sought ANC by treatment group and regions

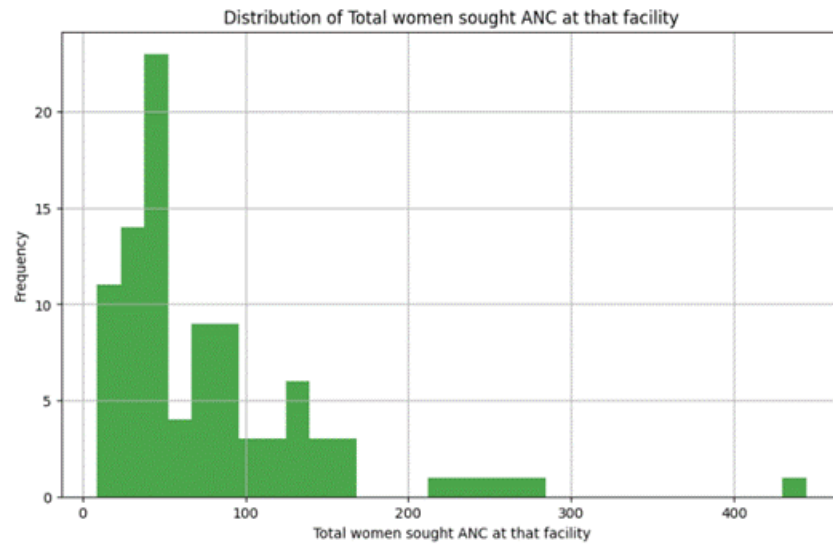


Figure 33. Histogram showing the distribution of total women who sought for ANC from selected participants.

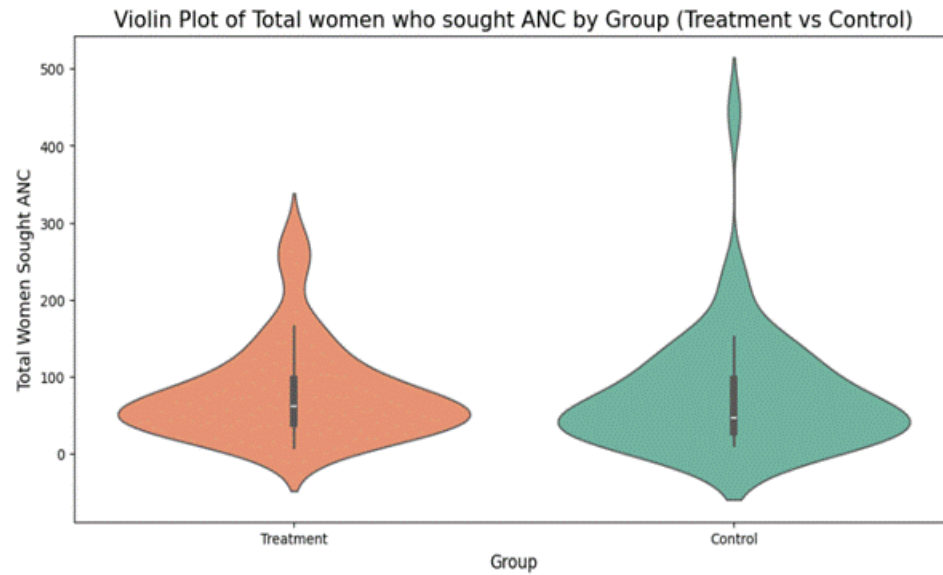


Figure 34. Violin plot showing the distribution of total women who sought for ANC by treatment group (Treatment vs Control).

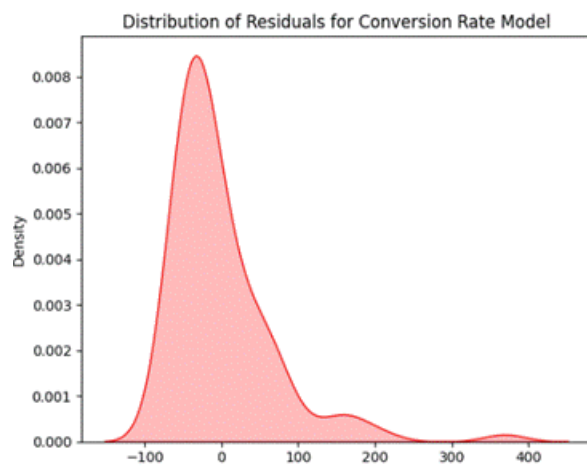


Figure 35. Density Plot

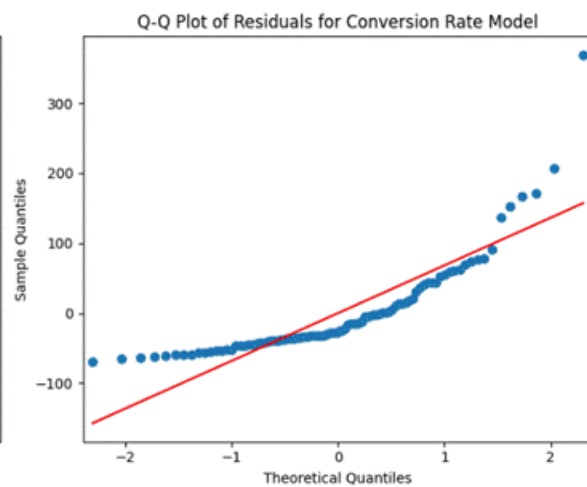


Figure 36. Q-Q Plot of Residuals

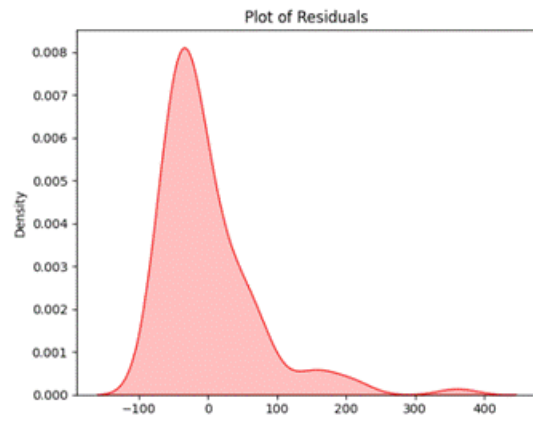


Figure 37. Density Plot

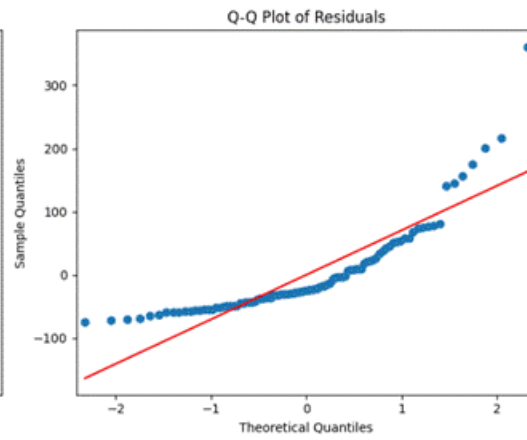


Figure 38. Q-Q Plot of Residuals

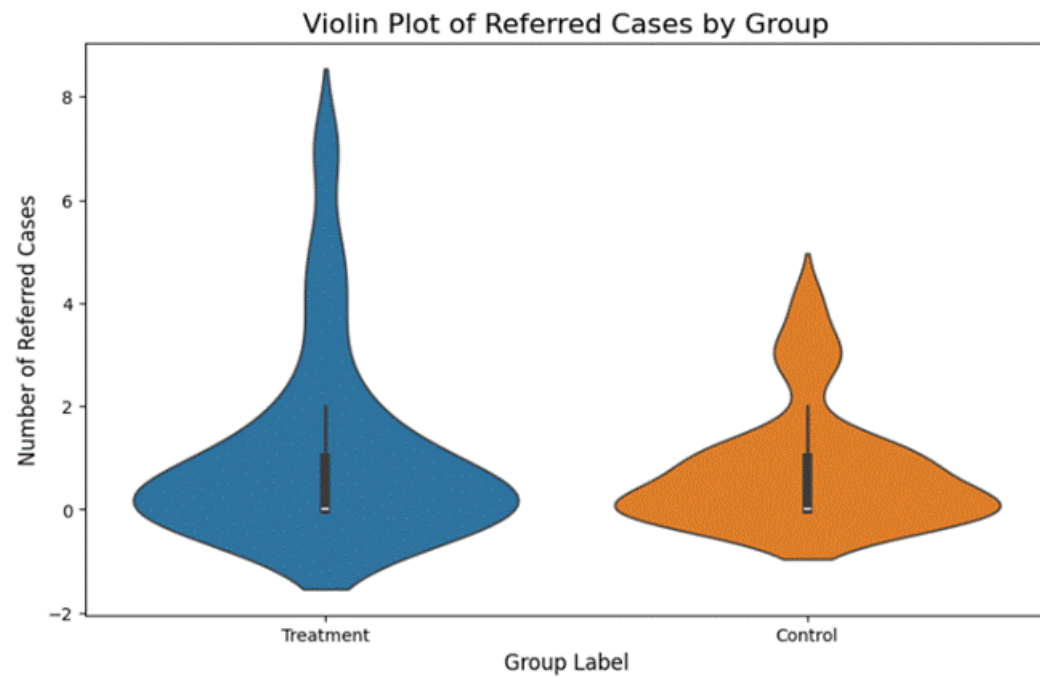


Figure 39. Violin plot comparing the distribution of number of referred cases in the treatment and control groups

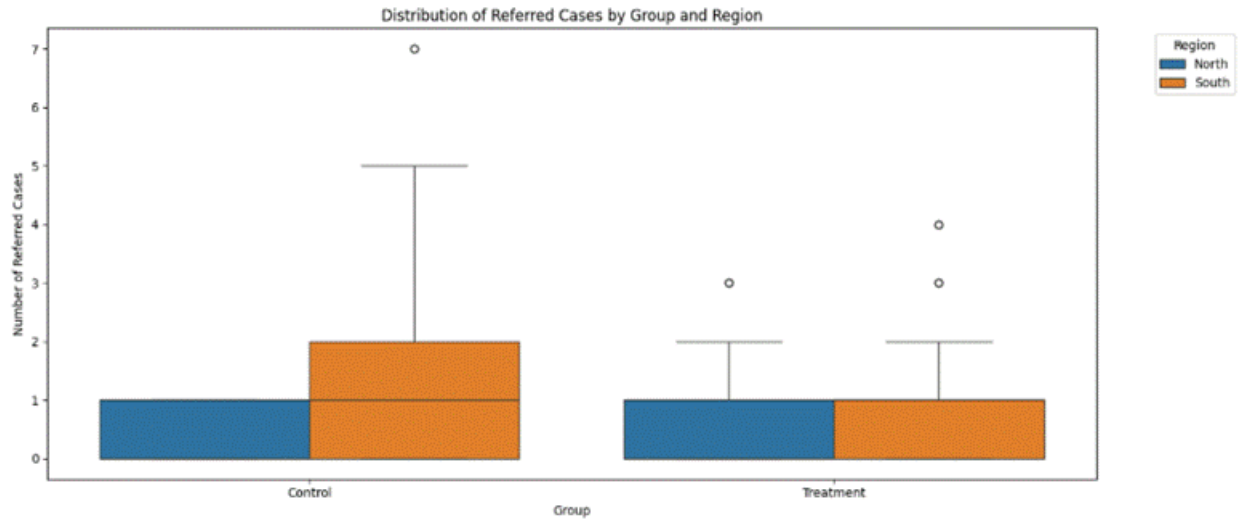


Figure 40. Boxplot showing the distribution of total number of women referred by treatment group and regions

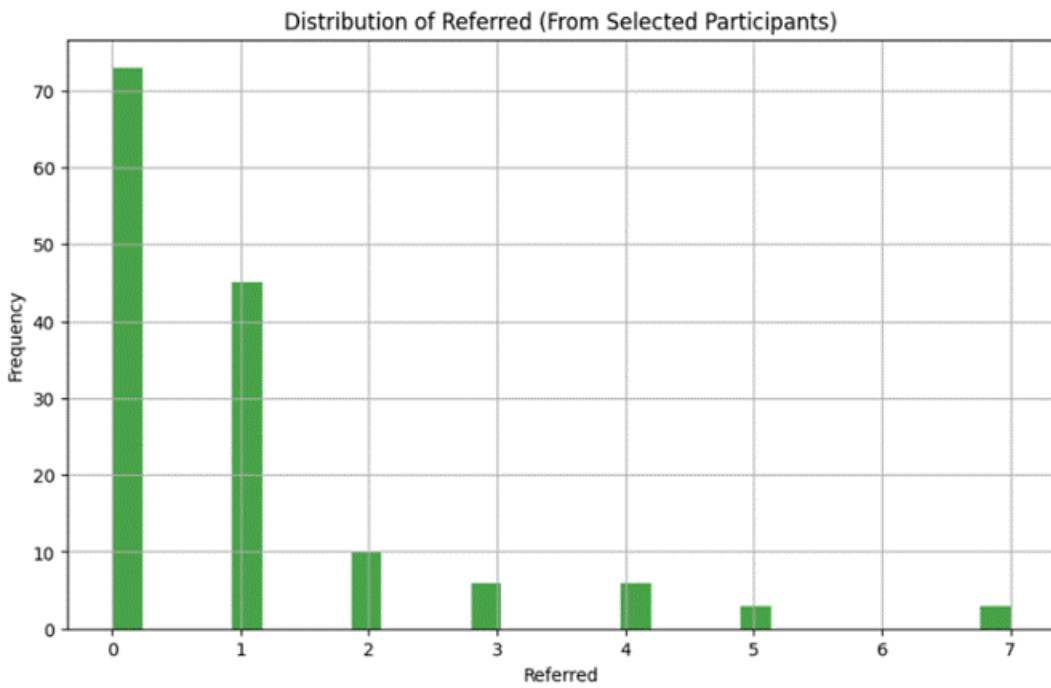


Figure 41. Histogram showing the distribution of total cases referred from selected participants.

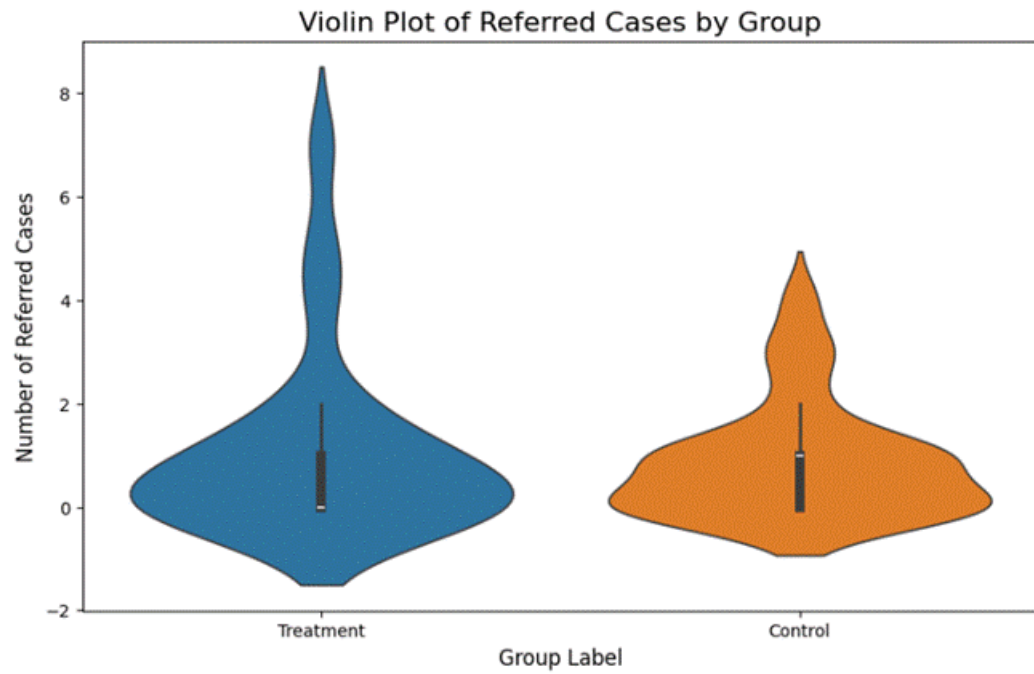


Figure 42. Violin plot showing the distribution of total cases referred after blocking (Treatment vs Control).

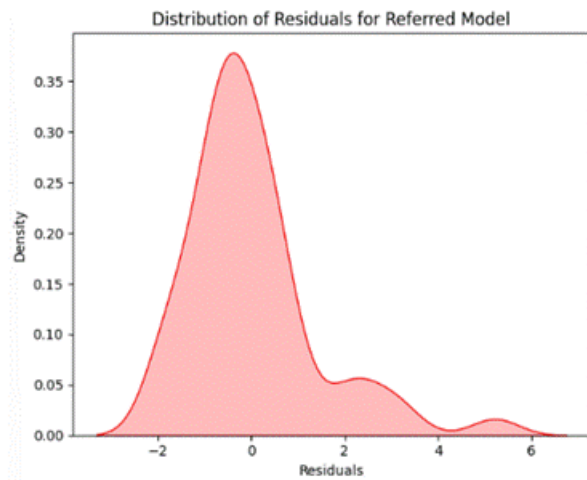


Figure 43. Density Plot

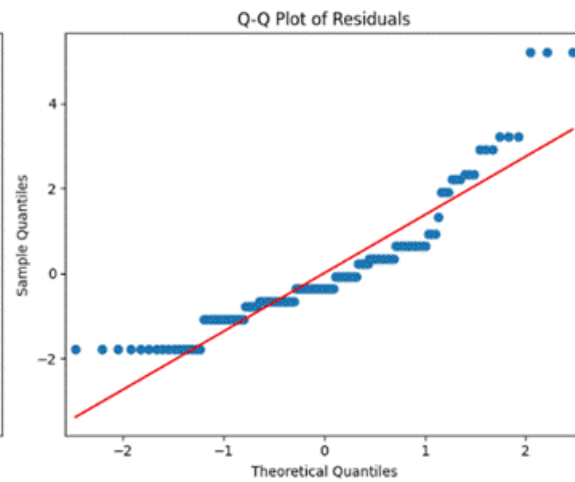


Figure 44. Q-Q Plot of Residuals

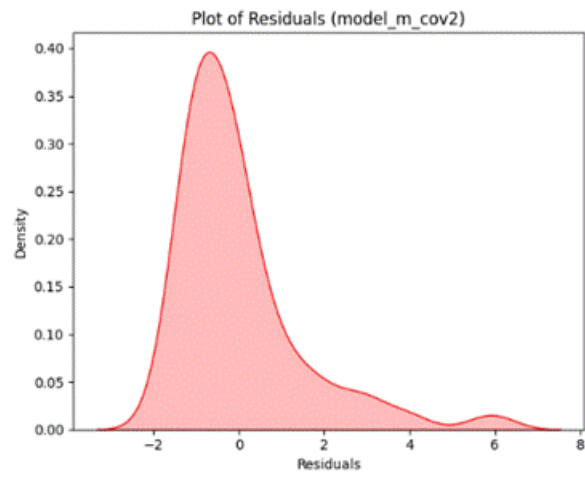


Figure 45. Density Plot

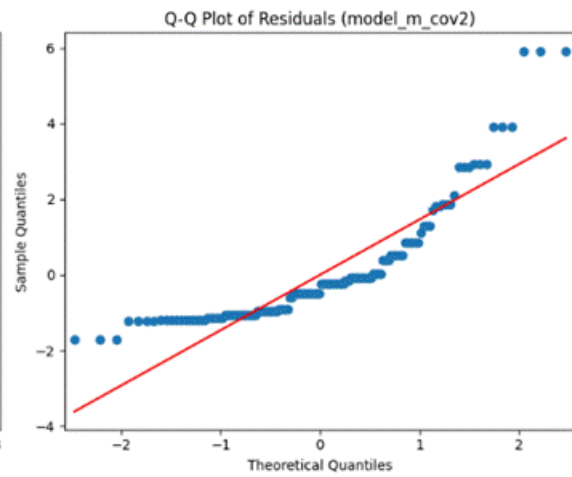


Figure 46. Q-Q Plot of Residuals